

MICROSOFT LEARN

Azure Cost Management

Portfolio writing sample

Introduction to analyzing costs and creating budgets with Microsoft Cost Management

Learn how to use cost analysis to understand how your costs accrue each month. Then use that understanding to create an Azure budget that monitors and alerts on your costs.

MODULE AT A GLANCE

Beginner

Administrator · Business analyst · Solution architect

Prerequisites

Owner access to an Azure subscription, five or more Azure resources with at least one month of cost history, and basic familiarity with the Azure portal.

7 units · Azure Cost Management

Authored and maintained by Bill Anderson during his Microsoft tenure

Source module

learn.microsoft.com/training/modules/analyze-costs-create-budgets-azure-cost-management/

MODULE OVERVIEW

Start with a cost-management question

In this scenario, you're a manager whose team has used Azure for several months. Your team has created multiple resources that cost money. You want to understand which resources drive the monthly bill and set thresholds that help your team stay within a finance-approved budget.

What you'll do

01 Set up

Apply planning, visibility, accountability, optimization, and iteration.

02 Analyze

Use Cost Analysis to identify the most expensive resources and spot trends.

03 Control

Create a budget and alert threshold based on what you found.

Learning objectives

- 1 Set up your organization for success when using Cost Management.
- 2 Analyze costs on a subscription using cost analysis.
- 3 Create your first Azure budget based on your analysis.

Before you begin

Confirm that you have Owner access to an Azure subscription, five or more Azure resources that have accrued cost for at least one month, and basic familiarity with the Azure portal.

UNIT 2 · COST MANAGEMENT PRINCIPLES

Build a repeatable cost practice

Microsoft Cost Management helps you plan for, analyze, and reduce spending so you can maximize your cloud investment. Use these principles to connect day-to-day resource decisions to business outcomes.

Planning

Ask what business problem you're solving and what usage patterns you expect before you create resources.

Visibility

Use Cost Analysis to see where money is spent and identify cost drivers and trends.

Accountability

Organize resources so you can attribute spending to the teams responsible for it.

Optimization

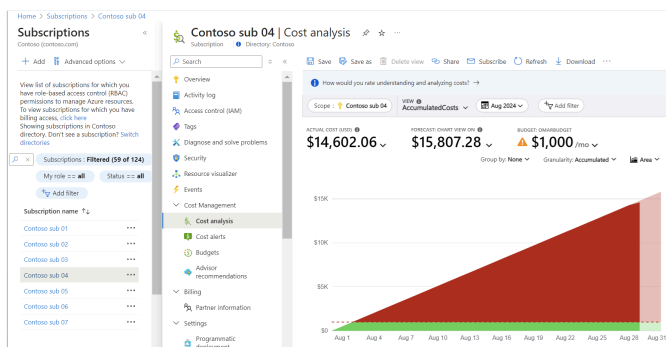
Use what you learn to improve purchasing, licensing, or infrastructure decisions.

Iteration

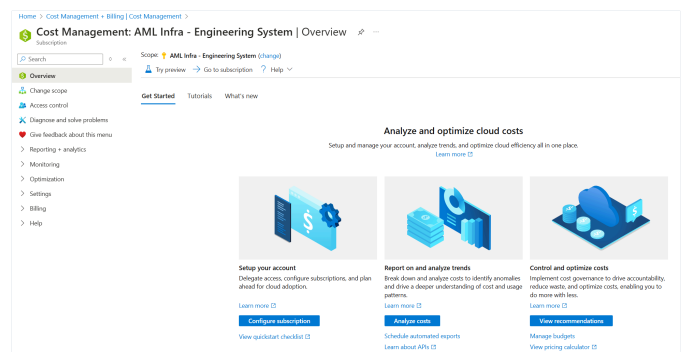
Keep teams engaged in an ongoing cycle of review, action, and optimization.

Where to find Cost Management

Access Cost Management tools from a subscription or resource group, or from the Cost Management + Billing window in the Azure portal.



Cost Management from a subscription or resource group.



Cost Management + Billing overview.

UNIT 3 · EVALUATE COSTS USING COST ANALYSIS

Read the story behind your spend

Cost Analysis gives you several views of cost and usage information. Start with the question you need to answer, then choose a date range, granularity, grouping, filter, and visualization that reveal the answer.

Accumulated cost

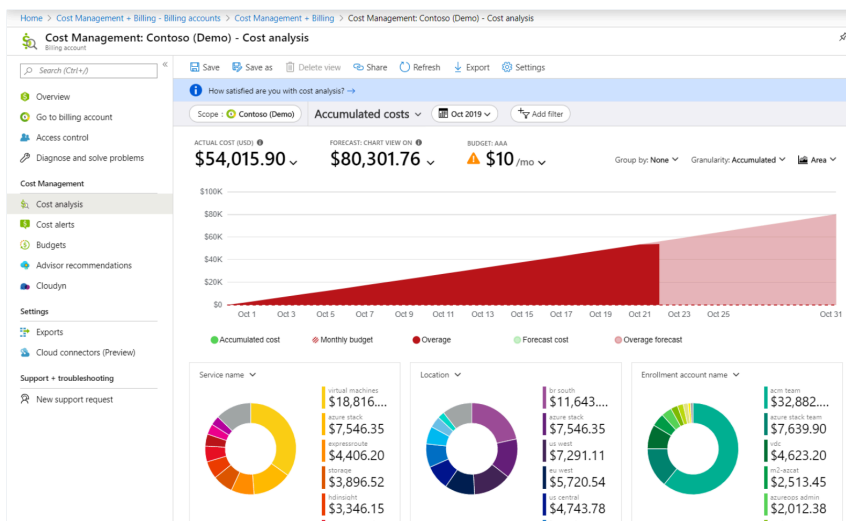
See how spending grows across the selected period and compare it with a budget.

Actual cost

See usage and purchase costs as they accrue and appear on your bill.

Forecast

Identify projected budget breaches and the estimated date of a breach.



A useful mental model

Cost Analysis is a set of views, not a single report. Change the view to match the decision you're making:

- How much have I spent?
- Which resources cost the most?
- How has spending changed over time?
- Am I on track to exceed the budget?

Cost by resource Aug 2019 Add filter

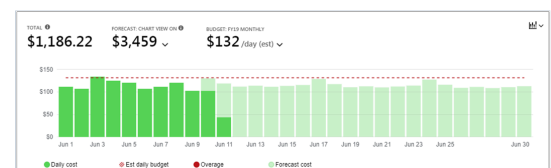
ACTUAL COST \$1,720.28 FORECAST: CHART VIEW ON BUDGET: MONTHLY \$1,500 /mo

Group by: Resource Granularity: None Table

Filter items 27 rows

Resource	Resource type	Location	Resource group name	Tags	Cost
month1stampfee_for_costanalysis	App Service Environment	us west	costmanagementdemo	Not applicable	\$684.43
month1stampfee_for_screenerenv	App Service Environment	us south central	webscreener	Not applicable	\$631.61
screenerfarm	App Service plan	us south central	webscreener	projectsilverscreen	\$132.60
data1	Disk	fr central	devops	Not applicable	\$93.81
appsvc_asp_linux_centralus	App Service plan	us central	appsvc_rg_linux_centralus	Not applicable	\$49.17
appservices	microsoft.security/pricings	unassigned	classic services	Not applicable	\$26.64
asp-costmanagementdemo-a13e	App Service plan	us west	costmanagementdemo	Not applicable	\$26.58
classiclab5851	Storage account (classic)	us central	classiclab	Not applicable	\$23.51

Cost by resource sorts resources from highest to lowest cost.



Daily costs show spikes and dips.

Use grouping and filtering to narrow the results.

EXERCISE 1 · SEE TRENDS

Find your most expensive resource

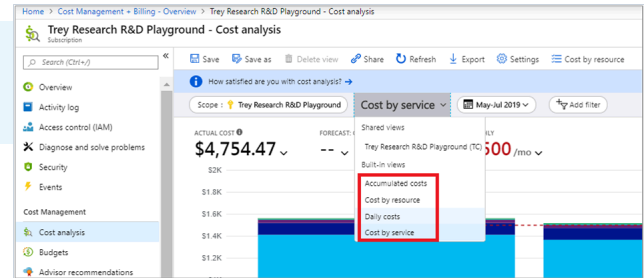
Select one of your Azure subscriptions, group the results by resource, and look for a trend over time. Record the resource name or resource ID. You'll use it in the budget exercise.

Exercise outcome

Identify the resource that contributes the most cost over the last three months.

Navigate to Cost Analysis

- 1 Sign in to the Azure portal.
- 2 In the portal search box, enter Subscriptions, then select Subscriptions.
- 3 Select a subscription.
- 4 Under Cost Management, select Cost analysis.



Choose a built-in view, then adjust its settings.

Group and adjust the results

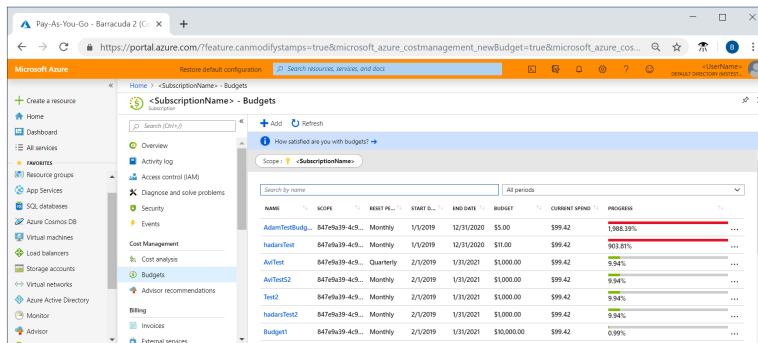
- 1 Select Group by, then select Resources.
- 2 Open the date list. Under Calendar months, select Last 3 months.
- 3 Note your most expensive resource.
- 4 Look for cost trends over time.
- 5 Record the resource name or resource ID for the next exercise.

Tip Actual cost is useful for reconciling your invoice. Amortized cost spreads reservation purchases across the reservation term, which can make trends easier to interpret.

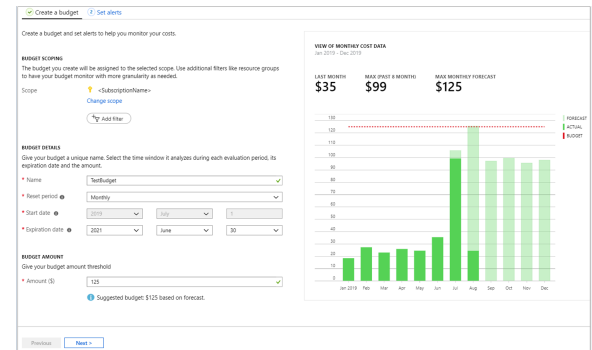
UNIT 5 · BUILD BUDGETS AND ALERTS

Turn analysis into an action

Create a budget at the scope that matches your decision. Your navigational context determines whether you create a budget for a subscription or a management group. Filters let you target specific costs, such as a resource group or service.



The Budgets pane lists existing budgets and provides Add.



The budget form shows a suggested amount and cost graph.

How budgets work

Scope

Choose the subscription or management group that owns the spending decision.

Filter

Apply a resource, resource group, or service filter to focus the budget.

Reset

Choose monthly, quarterly, annual, or billing-period evaluation.

Configure alerts

Budgets require at least one cost threshold and an email address. This module uses a 90 percent threshold. You can add up to five thresholds and five email addresses to a single budget. When a threshold is met, Cost Management generates an alert and sends email to the recipients.

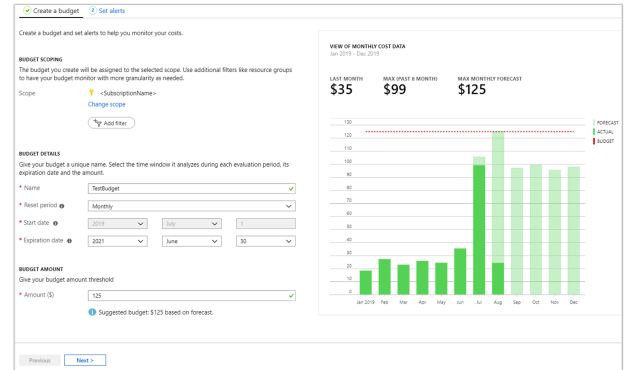
EXERCISE 2 · CREATE A BUDGET

Set a threshold for the cost driver

Use the resource you identified in the first exercise. Create a monthly budget for that resource, review the suggested amount and graph, and configure an alert at 90 percent.

Create and scope the budget

- 1 Open Subscriptions and select the subscription from the previous exercise.
- 2 Under Cost Management, select Budgets.
- 3 Select Add, then under Budget scoping select Add filter.
- 4 From the Filter list, select ResourceId.
- 5 Select the most expensive resource you identified.
- 6 Name the budget, for example MyFirstBudget.
- 7 Set Reset period to Monthly.



Review the suggested amount and monthly cost graph.

Review the recommendation

Examine the Budget Amount recommendation and the View of monthly cost data graph. Adjust the amount if it doesn't meet your needs. Select Next when you've decided on a budget amount.

Configure the alert

- 1 Set the alert condition to 90% of the budget. Don't specify an action group.
- 2 Enter your email address in Alert recipients (email).
- 3 Verify the graph and threshold.
- 4 Select Create.

SUMMARY

Keep the loop going

You started with a manager's question about Azure spending. You applied Cost Management principles, used Cost Analysis to understand the data, found the most expensive resource, and created a budget with an alert to keep costs under control.

A practical sequence

1**See**

Use Cost Analysis to understand where the money goes.

2**Decide**

Interpret trends and choose a threshold that fits the work.

3**Act**

Create a budget and alert the people responsible.

4**Iterate**

Review results and optimize resource use over time.

Portfolio context

Bill Anderson authored and maintained Microsoft Learn training for Azure Cost Management and related scenarios during his Microsoft tenure. His work included owning about five Cost Management learning modules, tracking completion rates, improving lower-performing modules, and creating instructional scripts, videos, and supporting learning assets. This flagship module was localized into 17 languages.

Learn more

[Quickstart: Start using Cost analysis](#)

[Trigger an action group with budgets](#)